

Scenario:

Dr. Robert Quincy is the chief Forensics Examiner at the UNCC Forensics Laboratory. He has directed you to perform technical work on a forensics image using lab equipment and software. Specifically, he would like you to verify a forensics image provided by the Cybersecurity Center at UNCC and determine, if you can, whether the images are of the same thumb drive. The University Attorney has determined that the original device was abandoned property, and no legal authority is required.

Forensic Acquisition & Exam Preparation:

Having booted the forensic workstation to Windows 11, I verified the forensic image using FTK Imager 8.2.0.

Following this, all request items are satisfied by the use of FTK Imager 8.2.0.

I recorded and verified hash values match and are as follows:

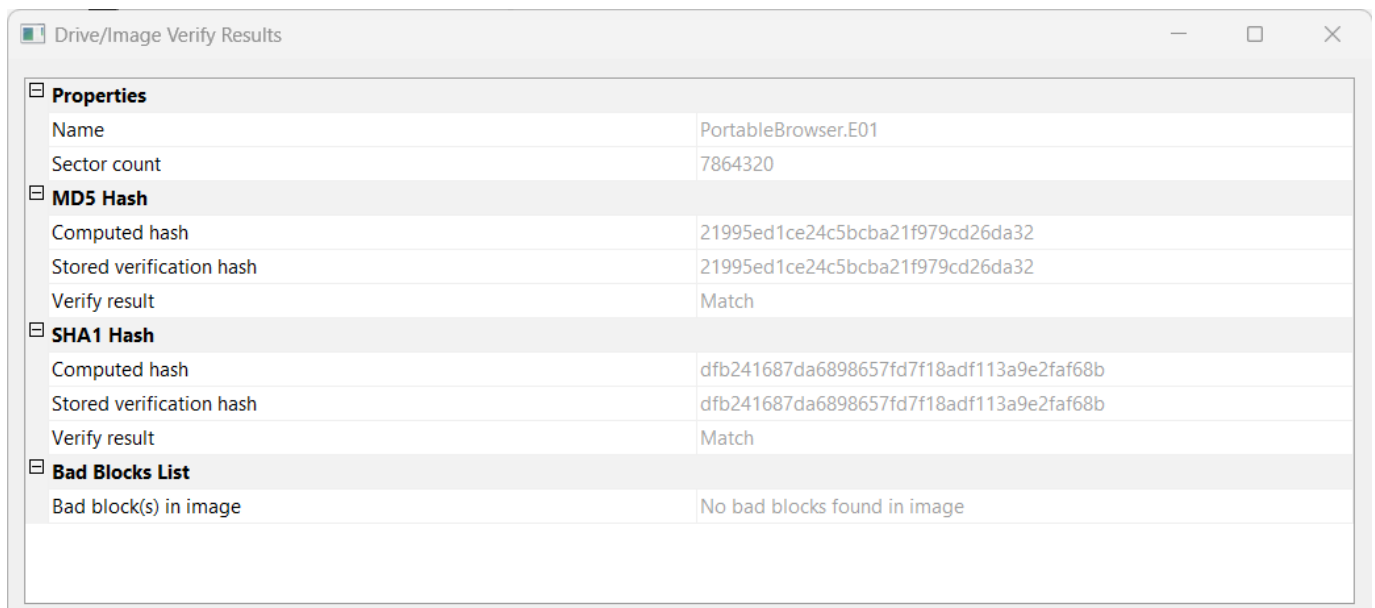


Figure 1: FTK Imager PortableBrowser.E01

Findings and Report (Forensic Analysis)

Using FTK imager, I answered all the questions asked by the referring agency.

Each item is addressed separately below.

a. What was the MD5 hash value for PortableBrowser.e01? How about CoffeeShopper?

By using FTK Imager, I verified the image as stated above. The MD5 hash value for PortableBrowser.e01 is 21995ed1ce24c5bcba21f979cd26da32. For CoffeeShopper the MD5 hash value is bc7583a1aede728b05df71f3120603c7.

b. What file systems are present within PortableBrowser.e01? (FAT32, NTFS, EXT3, Reiser, ZFS, UDF, ETC)?

FAT32 is the only file system present within PortableBrowser.e01. I found this while looking at the raw binary data within it I found FAT32 listed when FTK imager was listing how some of the disk was corrupted.

```
30000000 FA B8 00 00 8E D0 BC 00-7C 8B F4 50 07 50 1F FB d...Bx...pP-g
30000010 FC BF 00 06 B9 00 01 F3-A5 EA 1E 06 00 00 BE BE d...-...-N
30000020 07 80 3C 80 74 02 CD 18-56 53 06 BB 00 7C B9 01 <<t-i-VS-w-|+
30000030 00 BA 00 00 B8 01 02 CD-13 07 5B 5E B2 80 72 0B &...-i-[-*+
30000040 BF BC 7D 81 3D 55 53 75-02 B2 00 BF EB 06 88 15 &N)-USu-+g&-...
30000050 8A 74 01 8B 4C 02 8B EE-EB 15 BE 9B 06 AC 3C 00 t-L...ie-N...c-
30000060 74 0B 56 BB 07 00 B4 0E-CD 10 5E EB F0 EB FE BB t-Va...i-+e&hp
30000070 00 7C B8 01 02 CD 13 73-05 BE B3 06 EB DF BE D2 |...-i-g-+p-s&ND
30000080 06 BF FE 7D 81 3D 55 AA-75 D3 BF 24 7C BE EB 06 &b)-u*uo;g(Ne-
30000090 8A 04 88 05 8B F5 EA 00-7C 00 00 49 6E 76 61 6C ....de-|)-Inval
300000a0 69 64 20 70 61 72 74 69-74 69 6F 6E 20 74 61 62 id partition t&b
300000b0 6C 65 00 45 72 72 6F 72-20 6C 6F 61 64 69 6E 67 le-Error loading
300000c0 20 6F 70 65 72 61 74 69-6E 67 20 73 79 73 74 65 operating syste
300000d0 6D 00 4D 69 73 73 69 6E-67 20 6F 70 65 72 61 74 m.Missing operat
300000e0 69 6E 67 20 73 79 73 74-65 6D 00 00 00 00 00 00 ing system.....
300000f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000100 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000110 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000120 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000130 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000140 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000150 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000160 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000170 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000180 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000190 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300001a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300001b0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 04 .....
300001c0 05 00 0B 1E DE CF 00 01-00 00 00 FF 77 00 00 00 ...&l...-p-...
300001d0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300001e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300001f0 00 00 00 00 00 00 00 00-00 00 00 00 55 AA .....D*
30000200 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000210 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000220 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000230 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000240 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000250 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000260 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000270 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000280 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
30000290 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300002a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300002b0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
300002c0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
Cursor pos = 174; phy sec = 0
```

Figure 2: FTK Imager PortableBrowser.E01

raw binary data.

- c. **What are the file size for PortableBrowser.e01 and CoffeeShopper.e01, and what was the size of the original device (hard drive) that PortableBrowser.e01 is imaged from? How about CoffeeShopper?**

The file size for PortableBrowser.e01 and CoffeeShopper.e01 are both 3839 MB. On the original device (hard drive) PortableBrowser.e01 is 537 KB while CoffeeShopper is 125,108KB.

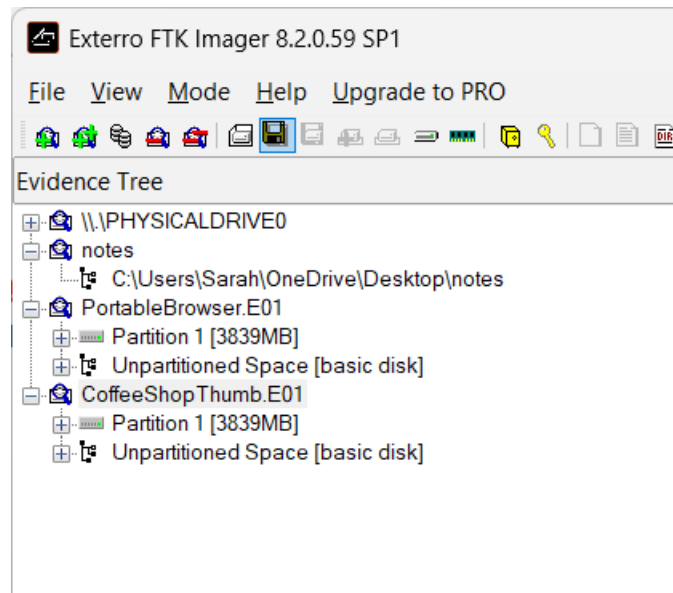


Figure 3: FTK Imager Volume sizes.

- d. **How are an image and a physical drive different as they appear in FTK Imager? (specifically, about the details of their appearance/labeling in FTK Imager)?**

As a physical drive you are allowed to see the manufactures information, it is also listed under physical drives, it also most importantly shows raw and unpartitioned views which if there are hidden areas they will become visible in this way. While as an image in FTK imager it is labeled with a filename/image file, it's path is also reflected of the image file on the disk, in this case it does show metadata that was embedded into the image like the examiner's name, case number, and acquisition date, which the raw physical drive cannot view these things.

- e. **Did FTK imager generate a log for the image you created out of PortableBrowser? Does this hash value match the original?**

Yes, FTK imager generated a log for the image I created out of PortableBrowser. The FTK image of PortableBrowser does match the original hash value once compared. I was able to find this answer when comparing the txt files to each other and matching each other when it came to their hash values.

f. **Do PortableBrowser and CoffeeShopper appear to be the same thumb drive? How are they similar and how are they different?**

They do appear to be on the same thumb drive, they are both Jetflash Transcend 4GB. On CoffeeShopper it states it in a different format stating it is a “4GB Transcend Jetflash Thumb drive (red and black)”. So, one of the major differences is it specifically stating that it is a red and black thumb drive.

Properties	
2 ↓	
Drive Geometry	
Bytes per Sector	512
Sector Count	7,864,320
Image	
Image Type	E01
Case number	CoffeeShopThumb
Evidence number	Item A
Examiner	Det. Peter Weller
Notes	Imaged in the UNCC Forensics Lab using a Wiebetec
Acquired on OS	Windows 7
Acquired using	ADI3.1.5.0
Acquire date	8/30/2016 9:27:55 PM
System date	8/30/2016 9:27:55 PM
Unique description	4GB Transcend Jetflash Thumb Drive (red and black)

Figure 4: FTK Imager Properties view showcasing CoffeeShopThumb has a unique description.

Conclusion

With this current analysis it is possible that these are two different thumb drives. There are parts of unpartitioned space for both and there is also the differential between both files as one is specifically stated to be a red and black one while also being phrased differently when it came to the type of thumb drive it was.

Signed:

Sarah Lopez Martinez

Student Sarah Lopez Martinez

Student Forensics Examiner

Appendix

File System: The organization of files and folders allocated to the user as they appear on a hard drive, thumb drive or camera card. File systems retain information about files like time, path, name and other meta information.

Hash value: A mathematical formula known as a Hash Algorithm can be applied to any piece of data to prove it is identical to another piece of data. Altering data by even a single bit will radically change its hash value.

Byte: 1024 bits (a bit is 1 or 0). Files, file systems and physical disks are represented by their size in bytes, kilobytes (KB), megabytes (MB), gigabytes (GB), terabytes (TB). Each order of bytes is comprised of 1000 of the previous order of bytes. Some systems count by the true "byte" value for 1000 instead, using orders of 1024. This number's significance is due to binary math, where increasing values of 1 or 0 are called "base 2". Thus, in order to count to 1024, 2^{10} bits are required.

Forensic Image: A file containing every bit from an original physical disk that can be compared by hash to be mathematically identical to the original item. Changing this image by even 1 bit will cause the hash values to be different.

Zone.Identifier: Is a value assigned in a Alternate Data Stream to a file in NTFS on Windows Operating systems to determine the source of a file on various networks.

Alternate Data Stream: An additional amount of data that is ad hoc sized for various purposes by either the Operating System or software and is not visible to the user and cannot be copied to non-NTFS file systems.

FAT32: A type of file system.